

- $\int \frac{1}{1+t^2} dt = \arctan t + C$
- $\int \frac{1}{\sqrt{1-t^2}} dt = \arcsin t + C$
- $\int \frac{1}{t} dt = \ln |t| + C$

- power reduction: $\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$
 $\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$
- double angle: $\sin 2\theta = 2 \sin \theta \cos \theta$
 $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$

- $\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$
- $\|\vec{v} \times \vec{w}\| = \|\vec{v}\| \|\vec{w}\| \sin \theta = [\text{parallelogram area}]$
- $|\vec{v} \cdot (\vec{w} \times \vec{r})| = [\text{parallelepiped volume}]$

- $\text{proj}_{\vec{v}}(\vec{w}) = \left(\frac{\vec{v} \cdot \vec{w}}{\vec{v} \cdot \vec{v}} \right) \vec{v}$
- $\text{comp}_{\vec{v}}(\vec{w}) = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\|}$

- $\text{distance}(Q, \mathcal{P}) = \frac{\|\vec{PQ} \cdot \vec{n}\|}{\|\vec{n}\|}$ where $\mathbf{P}$ is a point on $\mathcal{P}$
- $\text{distance}(Q, \ell) = \frac{\|\vec{PQ} \times \vec{v}\|}{\|\vec{v}\|}$ where $\mathbf{P}$ is a point on ℓ

- paraboloid: $z = x^2 + y^2$
- saddle: $z = x^2 - y^2$
- 1-sheeted hyperboloid: $x^2 + y^2 - z^2 = 1$
- 2-sheeted hyperboloid: $-x^2 - y^2 + z^2 = 1$
- ellipsoid—sphere if uniformly scaled:
 $x^2 + y^2 + z^2 = 1$
- double-cone: $z^2 = x^2 + y^2$

- tangent plane to the graph $z = f(x, y)$ at $P(a, b, \star)$ is:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$
- tangent plane to level set $F(x, y, z) = C$ at point P is:

$$\vec{\nabla} F(P) \cdot \vec{x} = \vec{\nabla} F(P) \cdot \vec{p}$$

- $D_{\vec{u}}f(P) = \vec{\nabla}f(P) \cdot \vec{u} = \|\vec{\nabla}f(P)\| \cos \theta$

- If $\mathbf{P}$ is a critical point of $f(x, y)$ and:
 $D = f_{xx}(\mathbf{P})f_{yy}(\mathbf{P}) - (f_{xy}(\mathbf{P}))^2$
 $T = f_{xx}(\mathbf{P}) + f_{yy}(\mathbf{P})$

$D < 0 \implies$ saddle
 $D > 0$ and $T > 0 \implies$ local minimizer
 $D > 0$ and $T < 0 \implies$ local maximizer

- An extremizer $\mathbf{P}$ of f subject to $g = C$ satisfies:
 $\vec{\nabla} f(\mathbf{P}) = \lambda \vec{\nabla} g(\mathbf{P})$ or $\vec{\nabla} g(\mathbf{P}) = \vec{0}$

- $r^2 = x^2 + y^2$ and $\tan \theta = \frac{y}{x}$
- $x = r \cos \theta$ and $y = r \sin \theta$
- $dA = r \, dr \, d\theta$ and $dV = r \, dz \, dr \, d\theta$